Mouad Imam

Software Engineer | PhD Candidate in Machine Learning

imam6mouad@gmail.com | +212 638 684 670 | immouad.com | GitHub | RubyGems | LinkedIn | Casablanca, Morocco

Summary

Software engineer with 3+ years of Ruby on Rails experience. I work with users and teammates to turn product needs into features, from backend APIs to frontend interfaces. I stay involved through testing and release, then improve the product based on feedback.

Professional Experience

Software Engineer
Jan 2026 – Present

Myunisoft (GeFi) · E-invoicing SaaS — *Remote from Morocco*

- Contributed to the Rails workflow for registering clients with France's accredited e-invoicing platform (PA), including testing at multiple layers.
- Built Rails and Angular features for invoice-address selection and status tracking, following Peppol and French e-invoicing requirements.
- Worked on Sidekiq jobs to update company contacts from public registries and process legacy data.
- Investigated application issues across teams and coordinated fixes with the engineers responsible.
- Contributed authentication fixes and Kubernetes network-policy changes to restrict service exposure.

Software Engineer
Jul 2024 – Dec 2025

Timlead · Enterprise Resource Planning — *Morocco · Part-time until May 2025*

- Built Rails and React ticketing and field-service features for field agents and operations teams.
- Added reusable ticket templates with configurable fields and conditional workflows for admins.
- Built a React/Stimulus editor so Rails APIs and Word/PDF exports could share response data.
- Built offline response bundles with media and signatures, preserving field validation rules.
- Automated dispatch scheduling based on distance and workload, with calendar and route views.
- Added CSV/XLSX import and export with column mapping and validation, reducing support escalations.
- Cut CI builds from **15 to 4 minutes** with Docker. Added hot reloads and commit hooks.

Software Developer
Aug 2023 – May 2025

Mibtech · HR & Healthcare Software — *Ruby on Rails · Morocco*

Radiology Information System
Feb – May 2025

- Designed PostgreSQL models for patient, exam, and report workflows, and built the registration, ordering, and report-validation screens.
- Implemented role-based permissions to control access to patient and imaging data.
- Built a TCP interface for data exchange between the Rails RIS and the hospital information system (HIS).
- Integrated the Rails app with Orthanc PACS for C-FIND/C-STORE exchanges and DICOM Modality Worklist for exam scheduling.
- Used logs and DICOM events to trace device-communication failures and fix integration issues. Wrote deployment and integration notes for the Dockerized RIS/PACS services.

Humaneo · HR Management
Aug 2023 – May 2025

- Worked directly with HR to build leave-request and contract-renewal workflows in Rails.
- Built a Rails scheduler to assign employees to shifts automatically, checking for overlaps and applying night-shift limits and contract rules.
- Added approval and onboarding emails to prompt HR staff and managers when requests needed attention.
- Built a training portal with videos, quizzes, and completion tracking for employees.
- Refactored employee records to avoid duplicate data across departments, and added validation and clear errors to contract and leave forms.

Education & Doctoral Research

University of Hassan II · Casablanca, Morocco

PhD Candidate · Machine Learning 2025 – Present

Personalized Passenger Guidance During Public Transport Disruptions

A route that suits one passenger can become crowded when many receive the same advice. I study how guidance can reflect individual travel preferences while accounting for demand on shared services. I compare guidance strategies in simulated disruptions on a multimodal network, measuring journey times and crowding for passengers who receive advice and for others using the network.

Master's · Big Data & Cloud Computing 2023 – 2025

Bachelor's · Computer Science & Applied Mathematics 2020 – 2023

Selected Projects

ProdLoger · Log Monitoring

Feb – Jul 2025

Master's Thesis · Hassan II University · Rust, Python, Elasticsearch, PostgreSQL

- Developed a machine learning model to predict system failures from application logs.
- Built a Rust pipeline for real-time log processing, with Elasticsearch for search and PostgreSQL for metadata.
- Designed log preprocessing and tuned anomaly detection thresholds to reduce false positives.
- Used Prometheus and Grafana to track processing latency, throughput, and alert activity.
- Automated load tests in GitHub Actions to check for performance regressions in the Dockerized services.

University Budget Management Platform

Apr – Jul 2023

Final-Year Project · Hassan II University · Next.js, MySQL

- Co-developed a Next.js/MySQL platform for budget planning and allocation across university departments.
- Automated budget distribution using configurable rules based on staffing and teaching workloads.
- Built dashboards for finance staff and department heads to manage allocations and track remaining budgets.

Published Ruby Gems

2026

Ruby, Rails, Active Record

- **eav_field** lets Rails teams configure custom fields on existing models as requirements change. It validates typed values and supports queries through Active Record. Used by Timlead.
- **validrb** gives API developers one set of Ruby schemas to validate and coerce request data and generate JSON Schema and OpenAPI definitions.
- **piperb** lets developers add or change pipeline steps by declaring their dependencies. It works out the run order and retries failures with configurable backoff.

Employee Monitor · Desktop Agent & Rails API

Jan 2026

Team Project · Electron, TypeScript, Ruby on Rails

- Blurred sensitive content in desktop screenshots with AWS Lambda before storing them in Amazon S3.
- Added location sync with Haversine distance checks for geofence entry, exit, and out-of-zone alerts.

Mangel · Manga and Novel Reader

Feb – Jun 2024

Personal Project · Ruby on Rails, React, MySQL

- Built a searchable Rails catalogue with separate list and detail responses and Active Storage covers.
- Added chapter navigation in Next.js using previous and next chapter references from the Rails API.
- Ran RSpec request specs against MySQL in GitHub Actions using Docker Compose.

Technical Skills

Backend & APIs: Ruby, Ruby on Rails, REST APIs, Sidekiq, WebSockets, SQL, Bash
Frontend: React, TypeScript, JavaScript, Angular, Next.js, Hotwire, Stimulus
Databases: PostgreSQL, MySQL, Redis, Elasticsearch
Infrastructure: Docker, Kubernetes, AWS, Google Cloud, Linux
Testing & CI: RSpec, Capybara, Jest, Cypress, Git, GitHub Actions, CI/CD
Monitoring: Sentry, Prometheus, Grafana, New Relic
Research: Python, Rust, PyTorch, TensorFlow, NLP, model training and evaluation
AI coding tools: Claude Code and Codex.